

SignWorld: A Unified AI Foundation for Saudi Sign Language Recognition & Generation

The Vision

Saudi Arabia is home to hundreds of thousands of Deaf and hard-of-hearing citizens, yet SSL remains one of the most under-resourced languages in the Kingdom's digital transformation. There is no unified, national-grade AI system that understands it, speaks it, or bridges it to the hearing world.

We are not building a simple app. We are building an infrastructure; a national digital nervous system for sign language, similar to how SDAIA treats Arabic NLP.

Just as SDAIA has invested in Arabic large language models as sovereign infrastructure, **SignWorld** proposes the same treatment for Saudi Sign Language: a single foundation model, "the Brain", that every future application, hospital, school, ministry, and developer in the Kingdom can build on top of, instead of everyone reinventing fragmented, low-quality sign language tools from scratch.

This aligns directly with **Vision 2030's** Quality of Life Program and disability inclusion mandates, and positions Saudi Arabia as the **first nation in the world with a state-hosted, foundation-model-grade sign language AI platform.**

The Core: SignWorld Brain

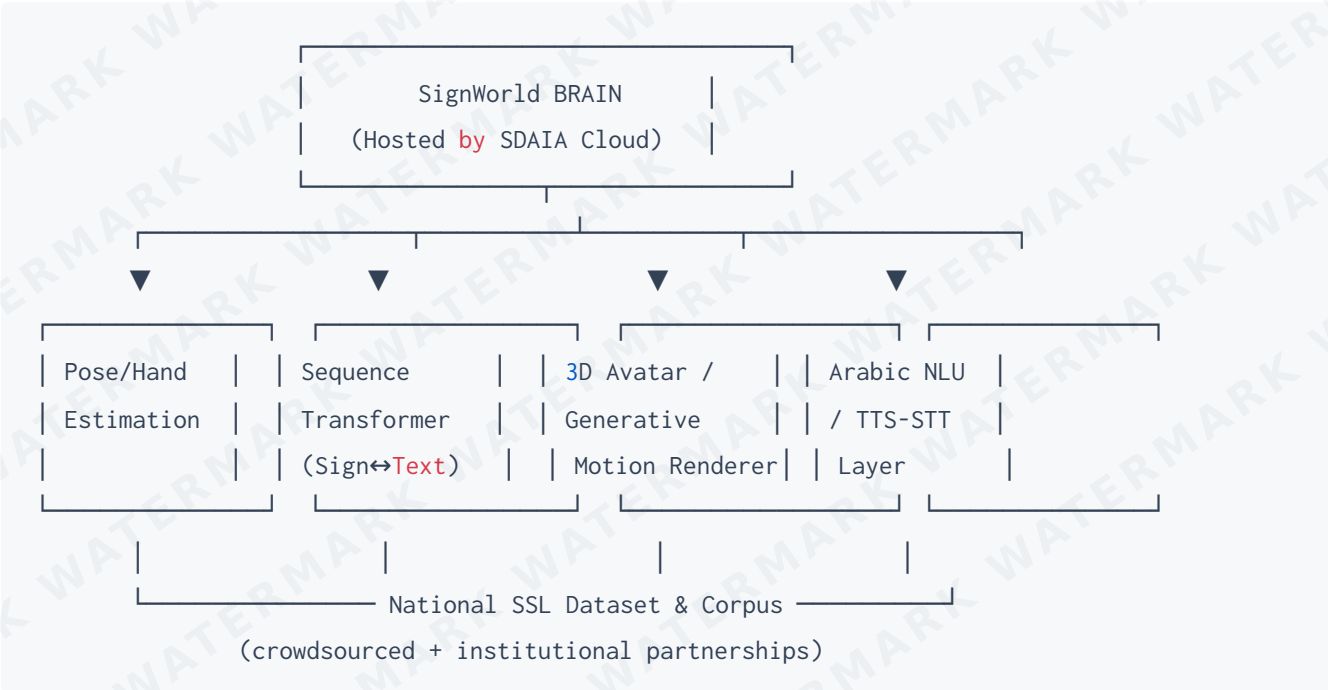
A single multimodal foundation model, conceptually hosted and governed by **SDAIA**, offered as a national API layer: **"Sign Language as a Service" (SLaaS).**

Four Core Capabilities

Capability	Direction	Description
Sign → Text/Speech	Recognition	Real-time video of a signer is translated into Arabic text or synthesized speech
Text/Speech → Sign	Generation	Arabic text or speech is converted into a photorealistic/avatar-based signing sequence

Capability	Direction	Description
Sign → Sign	Translation	SSL translated to/from other sign languages (ASL, international sign) – enabling Hajj/Umrah accessibility for millions of Deaf pilgrims worldwide
Sign → Sign (Dialect Normalization)	Standardization	Regional/International signing variants normalized into standard SSL, aiding consistency across the Kingdom

Conceptual Architecture



Technical Building Blocks:

- Brain that enables Sign2Text, Tex2Sign and Sign2Sign
- Client Applications
- For the hackathon demo: we need a **working thin slice** that *proves the architecture*, wrapped in a marketing of what it scales to.

Three Flagship Applications — Powered by One Brain

The genius of the pitch: **one core engine, three faces**. Each app is a thin, purpose-built UI calling the same SignWorld API — proving the platform's versatility live, on stage.

Education (Madrasati or something)

A real-time AI sign language interpreter for every classroom in Saudi Arabia.

- Teacher speaks → an on-screen avatar signs in real time for Deaf students, in any classroom, without needing a dedicated human interpreter present
- Deaf students can sign a question → converted instantly to text/speech for the teacher
- Integrates conceptually with the **Madrasati** platform (the national e-learning system), instantly reaching every public school in the Kingdom
- **Impact line:** *"Every classroom in Saudi Arabia, instantly accessible; no interpreter shortage can ever again stand between a Deaf child and their education."*

Medical

Breaking the silence in the emergency room.

- Hospital kiosk / tablet app for clinics and ERs: patient signs their symptoms → doctor receives real-time Arabic translation; doctor speaks → patient receives signed response from an avatar
- Critical for **emergency triage**, where waiting for a human interpreter can cost lives
- Deployable via **Seha app / MOH** integration as a long-term vision
- **Impact line:** *"In an emergency, seconds save lives. SignWorld makes sure no Deaf patient ever has to wait for someone to understand them."*

Children

Making sign language a joyful first language, not a barrier.

- Gamified app where Deaf children **and their hearing parents** learn SSL together through an animated avatar companion, mini-games, and story-based lessons
- Solves a heartbreaking real problem: **X%+ of Deaf children are born to hearing parents** who don't know sign language.
- Uses the *generation* side of the brain to produce endless new signed stories/content on demand
- **Impact line:** *"The first years of a child's life shape everything. SignWorld makes sure a hearing parent and their Deaf child never have to struggle to say 'I love you' to each other."*

Public Safety & National Services

Sign language for national emergency and government services.

- Live avatar-based sign interpretation integrated into **911/937 emergency calls**, so a Deaf caller can text-chat or sign via phone camera and be instantly understood by

dispatch

- Extends to **government broadcasts**: MOH announcements, weather emergencies, and national addresses automatically signed in real time by an AI avatar, something currently done manually and inconsistently on Saudi TV
- Extends further to **Hajj & Umrah accessibility**: real-time multilingual sign translation for Deaf pilgrims from around the world